

Additional file 1

ms-synth: a known-truth generator of synthetic longitudinal multiple sclerosis cohorts for benchmarking disease-progression methods

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All values are generated by the repository scripts (`analyze_manuscript_stats.py`, `benchmark_estimators.py`) and stored in `results/results.json` and `results/benchmark_summary.csv`.

Table S1: Per-transition recovery of the generating intensities in the reference cohort (N = 500, seed 42; heterogeneous population, relapse-triggered visits). Estimates in yr^{-1} ; relative bias (%) in parentheses. Asterisks mark relapse-active states.

Edge	States	True	Oracle	Crude	Panel likelihood
0→1	mild→mild*	0.60	0.326 (-45.7)	0.282 (-53.0)	0.386 (-35.6)
0→2	mild→mod	0.08	0.063 (-21.1)	0.047 (-41.0)	0.053 (-34.1)
1→0	mild*→mild	0.85	0.475 (-44.2)	0.344 (-59.5)	0.499 (-41.3)
1→2	mild*→mod	0.05	0.038 (-24.2)	0.037 (-25.6)	0.035 (-30.4)
1→3	mild*→mod*	0.10	0.080 (-20.4)	0.069 (-31.3)	0.092 (-8.3)
2→0	mod→mild	0.04	0.045 (+13.7)	0.036 (-11.0)	0.040 (-0.3)
2→3	mod→mod*	0.45	0.263 (-41.6)	0.221 (-50.8)	0.288 (-35.9)
2→4	mod→adv	0.07	0.078 (+11.7)	0.068 (-2.7)	0.078 (+12.0)
3→1	mod*→mild*	0.06	0.057 (-5.0)	0.046 (-23.2)	0.058 (-3.2)
3→2	mod*→mod	0.60	0.336 (-44.1)	0.256 (-57.4)	0.352 (-41.3)
3→4	mod*→adv	0.05	0.051 (+1.4)	0.046 (-7.8)	0.043 (-13.4)
3→5	mod*→adv*	0.08	0.071 (-11.4)	0.064 (-20.1)	0.076 (-4.6)
4→5	adv→adv*	0.30	0.209 (-30.2)	0.187 (-37.6)	0.231 (-23.1)
4→7	adv→early	0.06	0.059 (-2.2)	0.050 (-16.8)	0.056 (-6.7)
5→4	adv*→adv	0.40	0.305 (-23.9)	0.241 (-39.6)	0.318 (-20.4)
5→6	adv*→early*	0.08	0.085 (+5.9)	0.073 (-9.3)	0.087 (+9.1)
5→7	adv*→early	0.04	0.047 (+18.0)	0.049 (+22.2)	0.045 (+13.2)
6→7	early*→early	0.50	0.363 (-27.5)	0.247 (-50.6)	0.346 (-30.7)
6→8	early*→late*	0.10	0.125 (+24.7)	0.137 (+37.2)	0.146 (+45.8)
7→6	early→early*	0.15	0.112 (-25.2)	0.097 (-35.5)	0.121 (-19.4)
7→8	early→late*	0.12	0.146 (+21.6)	0.131 (+8.9)	0.142 (+18.6)
8→9	late*→late	0.35	0.288 (-17.7)	0.226 (-35.6)	0.273 (-22.0)
8→10	late*→severe	0.08	0.158 (+97.6)	0.143 (+78.7)	0.149 (+86.4)
9→8	late→late*	0.08	0.090 (+12.1)	0.079 (-1.6)	0.105 (+31.2)
9→10	late→severe	0.10	0.127 (+27.1)	0.122 (+22.4)	0.127 (+27.0)
10→11	severe→abs.	0.05	0.029 (-42.2)	0.033 (-34.2)	0.031 (-38.2)
Mean relative error			0.254	0.313	0.251
Mean relative bias			-7.4%	-18.2%	-6.4%

Table S2: Full results of the estimator benchmark. Design A: cohort size with relapse-triggered visits; B: visit frequency (interval multiplier) with non-informative visits; C: cohort size with non-informative visits. Bias: mean relative bias over the 26 intensities (%), Monte Carlo SE in parentheses; |err|: mean absolute relative error (MCSE). Five replicates (seeds 1000–1004) per row.

Design	Population	Relapse visits	Interval ×	N	Estimator	Bias (%)	err
A	heter.	yes	1	250	crude	-20.3 (2.0)	0.352 (0.016)
A	heter.	yes	1	250	oracle	-8.8 (2.1)	0.308 (0.016)
A	heter.	yes	1	250	panel	-6.7 (2.5)	0.317 (0.018)
A	heter.	yes	1	500	crude	-21.4 (1.7)	0.317 (0.013)
A	heter.	yes	1	500	oracle	-10.2 (2.0)	0.274 (0.010)
A	heter.	yes	1	500	panel	-8.9 (2.2)	0.272 (0.013)
A	heter.	yes	1	1000	crude	-20.3 (1.0)	0.314 (0.007)
A	heter.	yes	1	1000	oracle	-9.3 (1.2)	0.255 (0.006)

Design	Population	Relapse visits	Interval $\times$	N	Estimator	Bias (%)	err
A	heter.	yes	1	1000	panel	-7.9 (1.3)	0.258 (0.007)
A	heter.	yes	1	2000	crude	-20.6 (0.5)	0.313 (0.006)
A	heter.	yes	1	2000	oracle	-9.8 (0.7)	0.241 (0.006)
A	heter.	yes	1	2000	panel	-8.2 (0.7)	0.239 (0.010)
A	heter.	yes	1	4000	crude	-20.6 (0.6)	0.310 (0.005)
A	heter.	yes	1	4000	oracle	-10.1 (0.7)	0.240 (0.003)
A	heter.	yes	1	4000	panel	-8.3 (0.7)	0.237 (0.006)
A	homog.	yes	1	250	crude	-13.4 (2.0)	0.241 (0.020)
A	homog.	yes	1	250	oracle	+2.1 (2.9)	0.164 (0.009)
A	homog.	yes	1	250	panel	+9.2 (3.2)	0.274 (0.018)
A	homog.	yes	1	500	crude	-14.8 (1.5)	0.202 (0.009)
A	homog.	yes	1	500	oracle	+1.1 (1.7)	0.117 (0.007)
A	homog.	yes	1	500	panel	+6.8 (1.6)	0.206 (0.004)
A	homog.	yes	1	1000	crude	-15.1 (1.5)	0.196 (0.006)
A	homog.	yes	1	1000	oracle	-0.1 (1.6)	0.085 (0.007)
A	homog.	yes	1	1000	panel	+5.7 (1.6)	0.169 (0.009)
A	homog.	yes	1	2000	crude	-14.9 (1.1)	0.170 (0.002)
A	homog.	yes	1	2000	oracle	-0.2 (1.2)	0.060 (0.005)
A	homog.	yes	1	2000	panel	+5.7 (1.3)	0.148 (0.008)
A	homog.	yes	1	4000	crude	-14.7 (0.6)	0.158 (0.003)
A	homog.	yes	1	4000	oracle	-0.0 (0.6)	0.037 (0.002)
A	homog.	yes	1	4000	panel	+5.8 (0.6)	0.131 (0.004)
B	homog.	no	0.5	1000	crude	-12.8 (1.4)	0.185 (0.008)
B	homog.	no	0.5	1000	oracle	+0.0 (1.2)	0.087 (0.008)
B	homog.	no	0.5	1000	panel	-0.1 (1.5)	0.103 (0.010)
B	homog.	no	2	1000	crude	-33.4 (1.5)	0.387 (0.011)
B	homog.	no	2	1000	oracle	-0.6 (1.3)	0.076 (0.005)
B	homog.	no	2	1000	panel	-0.3 (1.7)	0.125 (0.011)
C	homog.	no	1	250	crude	-19.6 (0.7)	0.289 (0.022)
C	homog.	no	1	250	oracle	+1.8 (1.2)	0.172 (0.016)
C	homog.	no	1	250	panel	+1.4 (1.4)	0.212 (0.030)
C	homog.	no	1	500	crude	-21.1 (1.1)	0.285 (0.008)
C	homog.	no	1	500	oracle	+1.1 (0.6)	0.127 (0.005)
C	homog.	no	1	500	panel	+0.1 (1.0)	0.160 (0.020)
C	homog.	no	1	1000	crude	-21.4 (0.8)	0.260 (0.007)
C	homog.	no	1	1000	oracle	+0.6 (0.7)	0.087 (0.004)
C	homog.	no	1	1000	panel	+0.0 (0.9)	0.104 (0.009)
C	homog.	no	1	2000	crude	-20.7 (0.7)	0.247 (0.002)
C	homog.	no	1	2000	oracle	+0.2 (0.6)	0.054 (0.004)
C	homog.	no	1	2000	panel	+0.4 (0.8)	0.069 (0.005)
C	homog.	no	1	4000	crude	-21.0 (0.6)	0.245 (0.004)
C	homog.	no	1	4000	oracle	+0.3 (0.2)	0.038 (0.002)
C	homog.	no	1	4000	panel	+0.1 (0.4)	0.049 (0.005)

Table S3: Inventory of the automated test suite (68 tests, all passing with ms-synth 1.0.0).

File	Test class	Tests	Scope
test_synthetic_data.py	TestBuildQFromConfig	7	shape, zero row sums, no ing state, entries
	TestScaleQForPatient	5	row sums after scaling; treatment effects
	TestSimulateTrajectory	8	state and EDSS ranges, 0 tone times, seeds
	TestGenerateDataset	19	schema, counts, EDSS, se ment, seeds, ages
	TestEpidemiologicalPlausibility	3	SPMS conversion, relapse up horizon
	TestValidationHelpers	2	empirical $\hat{\mathbf{Q}}$ and comparis
	TestHelperFunctions	9	grid snapping, state looku
	TestEdgeCases	5	single patient, short follow class cohorts
test_estimators_and_summaries.py	TestKnownTruthInterface	3	latent output leaves coho match visits; patient $\mathbf{Q}$ va
	TestEstimators	5	vectorised likelihood equ timator; oracle recovery; visit-rule switch
	TestSummaries	2	slope tolerance; headline ence cohort
Total		68	